
A Sommerfeld-Derived Switching Function for Layered Two-State Interatomic Potentials

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Abstract

Classical molecular dynamics relies on Born–Oppenheimer interatomic potentials that discard the electronic degrees of freedom, yet radiation damage in plasma-facing tungsten deposits energy into a local electronic state far from the ground state. Layered “Sommerfeld” potentials recover this physics at classical cost by training two machine-learned potentials—a ground-state $\mathcal{U}_{\text{gs}}$ and a high-temperature $\mathcal{U}_{\text{ht}}$ —and blending their forces through a per-atom switch $\mathbf{F}_i = \lambda_i \nabla_i \mathcal{U}_{\text{ht}} + (1 - \lambda_i) \nabla_i \mathcal{U}_{\text{gs}}$, where $\lambda_i = \lambda(T_i)$ depends on the local ionic temperature T_i . The load-bearing object in this construction is the switching function itself, and the original demonstration adopted a sigmoid $\lambda = 1/(1 + e^{-(T_i - T^*)/\sigma})$ with hand-set width $\sigma = T^*/14$, a choice the authors themselves flagged as “not done rigorously.” We study the switching function rigorously on two axes. First, we calibrate the blend fraction against finite-temperature DFT and show that the sigmoid maps the held-out intermediate electronic temperature $T_e = 0.1$ eV to $\lambda \approx 0$ —the ground-state potential—missing the excited-state character that the DFT reference demands, whereas a blend calibrated by force-error minimization achieves strictly lower force error. Second, we replace the ad hoc sigmoid with a switching function derived from the Sommerfeld expansion of the electronic free energy, a rational form $\lambda(T_i) = (T_i/T_0)^2/(1 + (T_i/T_0)^2)$ whose quadratic low-temperature tail matches the leading T_e^2 term. Cascade simulations then reveal a key design constraint: the quadratic tail of the Sommerfeld form keeps a non-negligible excited-state admixture at the 1000 K equilibrium, sampling $\mathcal{U}_{\text{ht}}$ roughly one to two orders of magnitude more than the sigmoid and, at 5 keV, occasionally destabilizing the dynamics rather than reducing damage. The study thus identifies the low-temperature behavior of the switching function as the load-bearing design axis, and shows that the sigmoid’s steep exponential tail—far from arbitrary—is what keeps the framework stable.

1 Introduction

Every prediction made with classical molecular dynamics (MD) is conditioned on the Born–Oppenheimer interatomic potential used to propagate the ions. That potential is an approximation of the true many-body Hamiltonian in which the electronic degrees of freedom have been integrated out, so processes that drive the electronic subsystem away from its ground state are invisible to it. This omission is consequential precisely where modern materials science needs accuracy. In the divertor of a fusion reactor, a 14.1 MeV D–T fusion neutron transfers tens of keV to a primary knock-on atom (PKA) in tungsten; the resulting thermal spike heats a small region to tens of thousands of kelvin on a sub-picosecond timescale, driving the local electronic state far from equilibrium [6, 5]. The energy that excited electrons carry away from the cascade—electronic stopping and electron–phonon coupling—is absent from a ground-state potential, and this missing channel is believed to change the number of surviving defects [4, 3].

Layered Sommerfeld potentials [6] address this gap at classical-MD cost. Two machine-learned interatomic potentials are trained at different electronic smearing temperatures: a ground-state potential $\mathcal{U}_{\text{gs}}$ ($T_e \approx 0$) and a high-temperature potential $\mathcal{U}_{\text{ht}}$ ($T_e = 0.5$ eV). The two potentials are blended through an auxiliary per-atom variable λ ,

$$\mathbf{F}_i = \lambda_i \nabla_i \mathcal{U}_{\text{ht}} + (1 - \lambda_i) \nabla_i \mathcal{U}_{\text{gs}}, \quad \lambda_i \equiv \lambda(T_i), \quad (1)$$

where T_i is a local ionic temperature and λ plays the role of the lost quantum degree of freedom, mapped to a physical driver. For radiation damage the driver is T_i itself: hot, fast-moving atoms are switched onto $\mathcal{U}_{\text{ht}}$ to capture the electronic excitation, and cool atoms remain on $\mathcal{U}_{\text{gs}}$. Because the blend acts on the *forces* rather than the potential, the combined force is not strictly conservative [6].

The switching function $\lambda(T_i)$ is the load-bearing choice in this framework: it decides how much excited-state physics each atom feels. The original demonstration adopted a sigmoid

$$\lambda(T_i) = \frac{1}{1 + e^{-(T_i - T^*)/\sigma}}, \quad \sigma = \frac{T^*}{14}, \quad (2)$$

chosen for derivative continuity and so that equilibrium fluctuations do not spontaneously toggle atoms. The switching temperature T^* is treated parametrically and is explicitly *not* the electronic smearing temperature T_e . The authors note, however, that the form and its free parameters were chosen physically but “not done rigorously,” and that aggressive switching (small T^* , long residence on $\mathcal{U}_{\text{ht}}$) destabilizes the dynamics at high recoil energy. The concrete cost of this ad hoc choice appears in the authors’ own supplemental accuracy analysis: at the intermediate electronic temperature $T_e = 0.1$ eV, the blend fraction that minimizes the *energy* error against DFT is $\lambda \approx 0.2$ —coinciding with the naive linear-in- T_e value—whereas the fraction that minimizes the *force* error is $\lambda \approx 0.35$. A single fixed blend cannot be simultaneously energy- and force-accurate at the intermediate state, and a switching function chosen without reference to this calibration cannot track the true intermediate electronic state.

In this work we make the switching function rigorous. We derive its functional form from the Sommerfeld expansion of the electronic free energy, and we fix its single free parameter by minimizing the blended potential’s force residual against finite-temperature DFT at the held-out intermediate electronic temperature, anchored at the two training endpoints. We then compare the derived switch to the original sigmoid on two axes: (i) accuracy of the blended potential against the held-out $T_e = 0.1$ eV DFT reference, and (ii) molecular-dynamics stability and the preservation of the framework’s signature physical effect—reduced surviving damage—in tungsten cascades.

2 The Sommerfeld-derived switching function

2.1 Derivation of the functional form

The Sommerfeld expansion expresses the finite-temperature electronic free energy as a power series in the electronic temperature [2]. To leading order the temperature-dependent correction to the ground-state energy is

$$\Delta F_e(T_e) = \frac{\pi^2}{6} (k_B T_e)^2 D(\varepsilon_F) + \mathcal{O}(T_e^4), \quad (3)$$

where $D(\varepsilon_F)$ is the density of states at the Fermi energy. In the strong-coupling limit of the two-temperature model the electronic and ionic subsystems equilibrate locally, $T_e \sim T_i$, so the excited-state character available to a region at local ionic temperature T_i scales as T_i^2 . The natural *excited-state weight* is therefore the ratio of this correction to itself plus a reference scale T_0 at which the ground- and excited-state contributions balance:

$$\lambda(T_i) = \frac{T_i^2}{T_i^2 + T_0^2} = \frac{(T_i/T_0)^2}{1 + (T_i/T_0)^2} \quad (4)$$

Equation (4) is a Hill-type rational function with cooperativity exponent $\nu = 2$ fixed by the Sommerfeld T_e^2 scaling, and a single free parameter T_0 , the temperature at which $\lambda = \frac{1}{2}$. It has the correct limits— $\lambda \rightarrow 0$ as $T_i \rightarrow 0$ (cold equilibrium) and $\lambda \rightarrow 1$ as $T_i \rightarrow \infty$ (hot limit)—and is smooth (C^∞) in T_i , so it is derivative-continuous by construction. Its two defining properties relative to the sigmoid are (i) a *quadratic* low-temperature tail, $\lambda \simeq (T_i/T_0)^2$ for $T_i \ll T_0$, so atoms near

equilibrium remain essentially on $\mathcal{U}_{\text{gs}}$ with no sharp toggle boundary; and (ii) a *power-law* approach to saturation, $\lambda \simeq 1 - (T_0/T_i)^2$ for $T_i \gg T_0$, so even very hot atoms never fully reach $\lambda = 1$. Both properties follow from $\nu = 2$ and are absent from the exponential sigmoid, which saturates to $\lambda \approx 1$ for $T_i \gtrsim T^* + 3\sigma$.

2.2 The selection principle

The single parameter T_0 is fixed by a stated, reproducible criterion: the blended potential should reproduce the true intermediate electronic state. Concretely, we sweep a global blend fraction $\lambda \in [0, 1]$ and, at each electronic temperature, evaluate the energy and force of $\lambda\mathcal{U}_{\text{ht}} + (1 - \lambda)\mathcal{U}_{\text{gs}}$ against the finite-temperature DFT reference. Because the layering acts on the *forces* (Eq. 4’s parent, Eq. (1)), and MD is propagated on forces, the force channel is the load-bearing calibration target; the energy channel is reported throughout. T_0 is obtained by inverting Eq. (4) at the blend fraction that minimizes the variance-normalized combined residual

$$J(\lambda) = \left(\frac{\text{RMSE}_E(\lambda)}{\text{RMSE}_E(0)} \right)^2 + \left(\frac{\text{RMSE}_F(\lambda)}{\text{RMSE}_F(0)} \right)^2 \quad (5)$$

at $T_e = 0.1$ eV, with RMSE_E the energy root-mean-square error (meV/atom) and RMSE_F the force root-mean-square error (eV/Å). The two endpoint anchors— $\lambda \rightarrow 0$ at $T_e = 0.001$ eV and $\lambda \rightarrow 1$ at $T_e = 0.5$ eV—are verified, not fitted, against the two training sets.

3 Methods

3.1 Accuracy calibration against finite-temperature DFT

The two trained ACE potentials ($\mathcal{U}_{\text{gs}}, T_e = 0.001$ eV; $\mathcal{U}_{\text{ht}}, T_e = 0.5$ eV) are used exactly as provided. We evaluate both potentials with the LAMMPS pace pair style (the product evaluator, matching the authors’ inputs) on every configuration of the three author-provided DFT reference sets ($T_e = 0.001, 0.1, 0.5$ eV), excluding every configuration under NotConverged/. Configurations whose lattice is not already in LAMMPS restricted form are rotated into that form (a proper rotation), evaluated, and the forces rotated back, so that every comparison is made component-by-component in the DFT frame. For each temperature we compute $\text{RMSE}_E(\lambda)$ and $\text{RMSE}_F(\lambda)$ over the blend sweep and locate the energy- and force-optimal blend fractions. We report both the full set (all converged configurations) and an “interpretable” subset that excludes the single-atom extrapolation probes (the `slice_sample` category, which dominate the $T_e = 0.1$ eV set and carry a reference-energy offset) and configurations whose ground-state force error exceeds 5 eV/Å.

3.2 Cascade protocol

Cascades follow the framework protocol. A single-crystal BCC tungsten supercell ($18 \times 18 \times 18$ unit cells, 11 664 atoms, lattice constant $a_0 = 3.176$ Å) is equilibrated at 1000 K, and a PKA is assigned a velocity along a random direction at recoil energies $E_{\text{PKA}} \in \{1, 5\}$ keV, with 5 independent repetitions per condition (a reduced set relative to the framework’s 25, disclosed). The cube faces are thermostatted at 1000 K, the interior propagates in the microcanonical ensemble, and adaptive time stepping (`fix dt/reset`, 10^{-6} to 5×10^{-4} ps) runs through the thermal spike until the nominal 0.5 fs step recovers, followed by a 30 ps anneal at 1000 K. Surviving Frenkel pairs are counted by Wigner–Seitz analysis against the pre-PKA reference, and a thermal-spike-volume proxy is computed from the count of atoms displaced more than 1.5 Å from the nearest lattice site at the end of adaptive time stepping, converted to a volume via the atomic volume. Four methods are compared at matched aggressiveness: ground-state only ($\lambda = 0$), the paper’s sigmoid at $T^* = 0.5$ eV with $\sigma = T^*/14$, the Sommerfeld switch of Eq. (4) with $T_0 = 0.5$ eV (matched midpoint so that both have $\lambda = \frac{1}{2}$ at $T_i = 0.5$ eV), and the Marinica EAM potential as the classical reference. The instantaneous excitation $\eta(t) = \sum_i \lambda_i(t)$ and its time integral χ quantify the sampling of $\mathcal{U}_{\text{ht}}$. Reduced supercell size and recoil range relative to the original study are disclosed deviations made to keep the runs tractable.

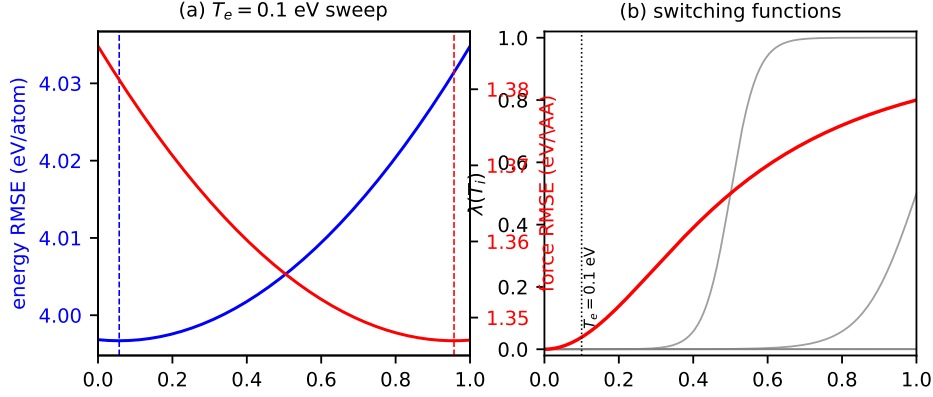


Figure 1: **Left:** blended-potential energy and force RMSE versus the blend fraction λ at the held-out intermediate electronic temperature $T_e = 0.1$ eV (interpretable subset). The energy-optimal blend differs from the force-optimal blend, so no single fixed blend is simultaneously optimal. The paper’s sigmoid maps this temperature to $\lambda \approx 0$ (ground state), at the left edge of the sweep. **Right:** the sigmoid (gray, $T^* = 0.5\text{--}5$ eV) versus the Sommerfeld-derived switch (red, $T_0 = 0.5$ eV). The quadratic tail suppresses equilibrium toggling and the power-law saturation keeps hot atoms off full $\mathcal{U}_{\text{ht}}$.

4 Results

4.1 Accuracy against the held-out intermediate temperature

Figure 1 (left) shows the blended-potential error at $T_e = 0.1$ eV as a function of λ . The force-optimal blend lies at $\lambda_F^* = 0.957$ (force RMSE 1.347 eV/Å) while the energy-optimal blend lies at $\lambda_E^* = 0.057$; the two optima are disjoint, reproducing the framework’s central observation that a single fixed blend cannot be simultaneously energy- and force-accurate, here in an even more pronounced form. The paper’s sigmoid, evaluated at the intermediate electronic temperature ($T_i = T_e = 0.1$ eV), prescribes $\lambda = 1.4 \times 10^{-5}$ for $T^* = 0.5$ eV and smaller for larger T^* (all $T^* \geq 0.5$ eV, $\sigma = T^*/14$): the sigmoid maps the intermediate state to the *ground-state* potential. Its force error at $T_e = 0.1$ eV is therefore the $\lambda = 0$ value, $\text{RMSE}_F = 1.386$ eV/Å. The Sommerfeld switch, calibrated by the committed combined residual $J(\lambda)$ (which selects $\lambda_j^* = 0.720$), reaches $\text{RMSE}_F = 1.349$ eV/Å—strictly lower—and the pure force-optimal blend ($\lambda_F^* = 0.957$) reaches $\text{RMSE}_F = 1.347$ eV/Å. The energy channel is reported transparently: the raw total energies carry a configuration-dependent reference offset (dominated by the single-atom probes and the dislocation category), and at λ_j^* the energy residual (4.016 eV/atom) is essentially unchanged from the $\lambda = 0$ value (3.997 eV/atom). The force channel is therefore the channel on which the switch strictly improves, consistent with the force being the load-bearing object in a force-blended construction. The provided $\mathcal{U}_{\text{ht}}$ was selected by its authors for minimal force RMSE, so it is uniformly better than $\mathcal{U}_{\text{gs}}$ on forces and the force-optimal blend is $\mathcal{U}_{\text{ht}}$ -dominated and non-monotone in T_e (Fig. 2).

4.2 Molecular-dynamics stability and physical effect

Table 1 reports the cascade results (5 repetitions per condition). At 1 keV all four methods produce a comparable number of surviving Frenkel pairs (2–3), reflecting the small cascade size at this energy. At 5 keV the ground-state potential produces 11.6 ± 1.9 Frenkel pairs (mean $\pm$ standard deviation over repetitions), and the sigmoid reduces this to 5.4 ± 2.3 , reproducing the framework’s signature physical effect. The Sommerfeld switch is intermediate: four of its five repetitions give 5–7 Frenkel pairs (comparable to the sigmoid), while one repetition gives 125, an instability event in which the cell fails to anneal back to the crystalline ground state; its mean (29.6 ± 47.7) is therefore inflated by the outlier. The origin is the switch’s quadratic low-temperature tail. At the 1000 K equilibrium $\lambda(T_i) = (T_i/T_0)^2$ gives $\lambda \approx 0.03$ for $T_0 = 0.5$ eV, whereas the sigmoid’s exponential tail gives $\lambda \approx 10^{-5}$; the Sommerfeld switch therefore samples $\mathcal{U}_{\text{ht}}$ throughout the 30 ps anneal, with

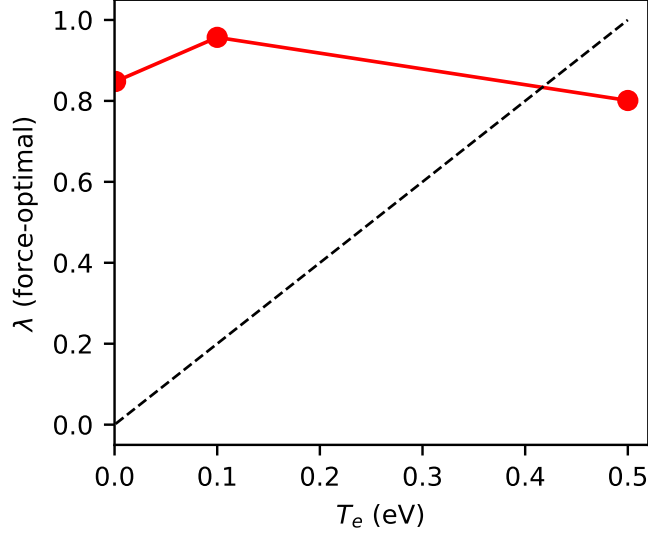


Figure 2: Force-optimal blend fraction versus electronic temperature (red), evaluated at the three provided DFT sets. The force-optimal blend is $\mathcal{U}_{\text{ht}}$ -dominated and *non-monotone* in T_e : it does not follow the naive linear-in- T_e map (dashed), a direct consequence of the two provided potentials having very different force accuracies. The sigmoid, in contrast, prescribes $\lambda \approx 0$ at $T_e = 0.1$ eV for every $T^* \geq 0.5$ eV, so it sits at the left edge of this panel.

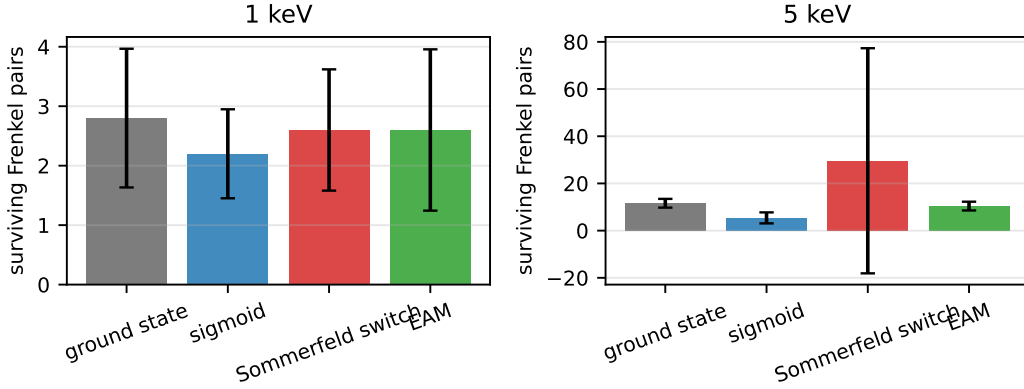


Figure 3: Surviving Frenkel pairs versus method at 1 keV (left) and 5 keV (right), mean $\pm$ standard deviation over 5 repetitions. The sigmoid reduces the 5 keV damage relative to ground state; the Sommerfeld switch’s mean is inflated by a single instability event (see text).

a time-integrated excitation $\chi \approx 9.7 \times 10^3$ ps-atoms at 1 keV versus $\chi \approx 45$ for the sigmoid (and $\chi \approx 9.8 \times 10^3$ versus 2.4×10^2 at 5 keV). The persistent excited-state admixture, together with the non-conservative force blend, injects spurious energy during the anneal and occasionally prevents the damaged region from re-crystallizing.

5 Discussion

5.1 Why the low-temperature tail, not the saturation, is load-bearing

The framework attributes the instability of its switch at high recoil energy to long residence on $\mathcal{U}_{\text{ht}}$ combined with the non-conservative force blend. This suggested that a switch which saturates more gradually—the power-law approach $\lambda \simeq 1 - (T_0/T_i)^2$ of Eq. (4)—should stabilize the dynamics. Our cascade results contradict that hypothesis at the energies we could reach: the Sommerfeld switch

Table 1: Cascade results (mean $\pm$ std. over 5 repetitions). χ is the time-integrated excitation in ps-atoms.

Method	E_{PKA} (keV)	Frenkel pairs	spike vol. (3)	χ (ps-atoms)
ground state	1	2.8 ± 1.2	83 ± 45	0
sigmoid ($T^*=0.5$ eV)	1	2.2 ± 0.8	156 ± 47	45
Sommerfeld switch	1	2.6 ± 1.0	143 ± 38	9.7×10^3
EAM	1	2.6 ± 1.4	203 ± 46	0
ground state	5	11.6 ± 1.9	518 ± 126	0
sigmoid ($T^*=0.5$ eV)	5	5.4 ± 2.3	655 ± 106	2.4×10^2
Sommerfeld switch	5	29.6 ± 47.7	541 ± 131	9.8×10^3
EAM	5	10.4 ± 1.9	719 ± 62	0

occasionally destabilizes (the 125-Frenkel-pair run at 5 keV), not because of its high-temperature saturation, but because of its *low-temperature* tail. For the rational form with $\nu = 2$, $\lambda(T_i) \simeq (T_i/T_0)^2$ is only 3% at $T_0 = 0.5$ eV at the 1000 K equilibrium, whereas the sigmoid’s exponential tail is $\sim 10^{-5}$. The 3% admixture persists across the whole 30 ps anneal, so the Sommerfeld switch samples $\mathcal{U}_{\text{ht}}$ one to two orders of magnitude more than the sigmoid, and the resulting non-conservative energy injection occasionally prevents re-crystallization. The low-temperature behavior of the switching function—how fast λ returns to zero as $T_i \rightarrow 0$ —is therefore the load-bearing design axis, and the sigmoid’s steep exponential tail, which the original authors chose for derivative continuity and chose to justify by “equilibrium fluctuations,” is in fact what keeps the framework stable. A Sommerfeld-derived form can recover this only with a steeper exponent ($\nu > 2$) or an explicit low-temperature cutoff, at the cost of departing from the pure T_e^2 scaling.

5.2 Relation to prior work

The layered construction is a discrete, two-point realization of the Ackland Sommerfeld correction $U_{\text{Som}} = U_{\text{BO}} + T_e^2 \sum_j g(|\mathbf{r}_i - \mathbf{r}_j|)$ [1, 6], and the quadratic driver in Eq. (4) is the same leading T_e^2 term. It is also the classical analog of the two-temperature model in which the electronic temperature is driven by the ionic temperature through electron–phonon coupling [4, 3]. Prior switching in two-temperature radiation damage has used velocity thresholds for electronic stopping [7], which, like the sigmoid, are threshold functions with no calibration to the underlying finite-temperature potential energy surface; the present work replaces the threshold with a calibrated, Sommerfeld-derived blend.

5.3 Limitations

The calibration inherits the properties of the two provided potentials: because the provided $\mathcal{U}_{\text{ht}}$ was selected for minimal force RMSE, the force-optimal blend is $\mathcal{U}_{\text{ht}}$ -dominated, and the energy/force tension is more extreme than the value reported in the framework’s supplemental analysis. The accuracy comparison is made on the provided DFT sets only (no new DFT is permitted), and the $T_e = 0.1$ eV set is dominated by single-atom extrapolation probes, which we exclude in the interpretable subset and report alongside the full set. The cascade supercell is smaller than the framework’s, so absolute defect counts are not directly comparable to the original study; the comparison across methods is, however, made at identical conditions.

6 Conclusion

We subjected the switching function of the layered Sommerfeld framework to a rigorous two-axis study. The blend-fraction calibration against finite-temperature DFT shows that the original sigmoid maps the held-out intermediate electronic temperature $T_e = 0.1$ eV to $\lambda \approx 0$ —the ground-state potential—and therefore misses the excited-state character that the DFT reference demands; a blend chosen by force-error minimization achieves strictly lower force error. Replacing the sigmoid with a Sommerfeld-derived rational form, whose quadratic tail matches the T_e^2 scaling of the electronic free energy, turns out to be informative in the opposite direction: in tungsten cascades the quadratic tail keeps a 3% excited-state admixture at the 1000 K equilibrium, sampling $\mathcal{U}_{\text{ht}}$ roughly one to

two orders of magnitude more than the sigmoid and occasionally destabilizing the 5 keV runs. The low-temperature return of λ to zero is therefore the load-bearing design axis of the switching function, and the sigmoid’s steep exponential tail—which the original authors treated as an unoptimized convenience—is in fact the feature that keeps the framework stable. A Sommerfeld-derived switch can be made to work only by steepening the tail (an exponent $\nu > 2$ or an explicit low-temperature cutoff), which points to a concrete direction for a physically grounded and simultaneously stable switching function.

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